

Maxillary Dental Anatomy and Physiology

Endodontic and Periodontal



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KEYWORDS

- Maxillary premolars • Maxillary molars • Apical periodontitis • Pulp necrosis
- Maxillary sinus

KEY POINTS

- The maxillary sinus floor is formed by the alveolar process of the maxilla.
- Endodontic infections are characterized by mixed biofilms of anaerobic and proteolytic bacteria that can infect the root canal space and accessory anatomy of those teeth producing apical periodontitis.
- Understanding maxillary sinus anatomy and its relationship with the posterior maxillary dentition is crucial for diagnosing and managing this entity.

INTRODUCTION

Understanding the anatomy and physiology of the maxillary sinus is crucial for diagnosing and managing sinus-related diseases and conditions that impact oral health. This article aims to provide a comprehensive overview of the anatomy of the maxillary sinus floor and its relationship with the maxillary posterior teeth. The clinical implications of these findings are critical for both diagnostic and therapeutic approaches in oral surgery, maxillofacial radiology, and endodontics.

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ANATOMY OF THE MAXILLARY SINUS FLOOR

The maxillary sinus floor is formed by the alveolar process of the maxilla and can vary significantly in its proximity to the roots of posterior teeth. This part of the sinus wall is typically the thinnest, making it susceptible to perforations during surgical dental procedures. In many individuals, the floor of the maxillary sinus dips down between the roots of the molars and premolars, creating extensions known as antral projections or alveolar recesses. In some cases, the roots of the posterior teeth can even protrude into the sinus cavity being separated only by the thin mucosal membrane. The thickness of the bone separating the sinus floor from the tooth roots can vary widely, from less than a millimeter to several millimeters. This variability can increase the likelihood of odontogenic sinusitis (ODS) following endodontic infections.¹ The maxillary sinus floor varies significantly between individuals. It is influenced by age, dental apical health, and previous maxillary sinus or dental surgeries. As people age or lose posterior teeth, the sinus often pneumatizes, expanding inferiorly to approximate or surround the roots of the remaining teeth.

CLASSIFICATION OF THE MAXILLARY SINUS FLOOR

The intricate relationship between the floor of the maxillary sinus and the roots of the posterior maxillary teeth demands a thorough understanding and careful clinical assessment before surgical interventions in this area.² The maxillary sinus floor's relationship to the teeth can be classified based on its topographic anatomy relative to the roots of the posterior teeth in:

Type A: The sinus floor is located above the apex of the posterior teeth, providing a significant bone buffer that reduces the risk of dental procedures impacting the sinus.

Type B: The sinus floor aligns closely with the apices of 1 or more posterior teeth but does not dip below them. This type requires caution during endodontic and surgical procedures to avoid sinus mucosal disruption.

Type C: The sinus floor extends between the roots of the posterior teeth, often dipping below the level of the root tips. This configuration presents the highest risk for sinus complications related to dental procedures.

These classifications² are clinically significant as they help predict the complexity of surgical and dental procedures like root-end surgeries, root canal treatments, sinus lifts, implants, and extractions. Knowing the classification can guide the surgeon's approach, predict possible complications, and influence the choice of techniques and tools.

ANATOMY OF MAXILLARY PREMOLARS AND MOLARS

Premolars derive their name from their position anterior to molars in the permanent dentition. According to the universal numbering system, maxillary first and second premolars in the upper right side are designated as 4 and 5, and 12 and 13 if they are located on the left side. According to the World Health Organization (WHO) nomenclature, more widely used in European countries, upper right premolars are numbered 14 and 15, and upper left 24 and 25. The maxillary first premolar exhibits variations in root morphology, which can include 2 well-developed roots, 2 root projections that are partially fused, or a single root. While the majority of maxillary first premolars typically contain 2 root canals, a small percentage may possess 3 roots. The maxillary second premolar serves a complementary function to the first premolar and closely resembles it in appearance. Generally, the root of the second premolar is comparable in length to or slightly longer than that of the first premolar. Although

most maxillary second premolars have a single root and canal, the presence of 2 roots or a single root with 2 canals is also possible (Fig. 1). Von arx and colleagues³ performed an analysis of the relationship between maxillary premolar apices and the floor of the maxillary sinus using cone-beam computed tomography (CBCT). Their findings revealed that palatal roots of 1st premolars were located 1 mm closer to the sinus compared to buccal roots. Roots of 2nd premolars were, on average, positioned much closer to the sinus than roots of first premolars. Protrusion of roots into the maxillary sinus was rare in 1st premolars, and low in 2nd premolars. Sex, age, and absence of the 1st premolar failed to have a significant effect on the mean distance between premolar roots and the border of the maxillary sinus.³

According to the universal numbering system maxillary molars in the upper right area are designated as 1, 2, and 3 for the third, second, and maxillary first molars. In the upper left area, the maxillary first, second, and third molar are designated with the numbers 14, 15, and 16. Again, in the WHO nomenclature, upper right molars become 16, 17, and 18, and left molars 26, 27, and 28. The maxillary first molar typically stands as the largest tooth in the maxillary arch, featuring 3 roots and 4 canals. Among these, the palatal root typically has the greatest dimensions, followed by the mesiobuccal (MB) and distobuccal roots. The MB roots often harbors an additional

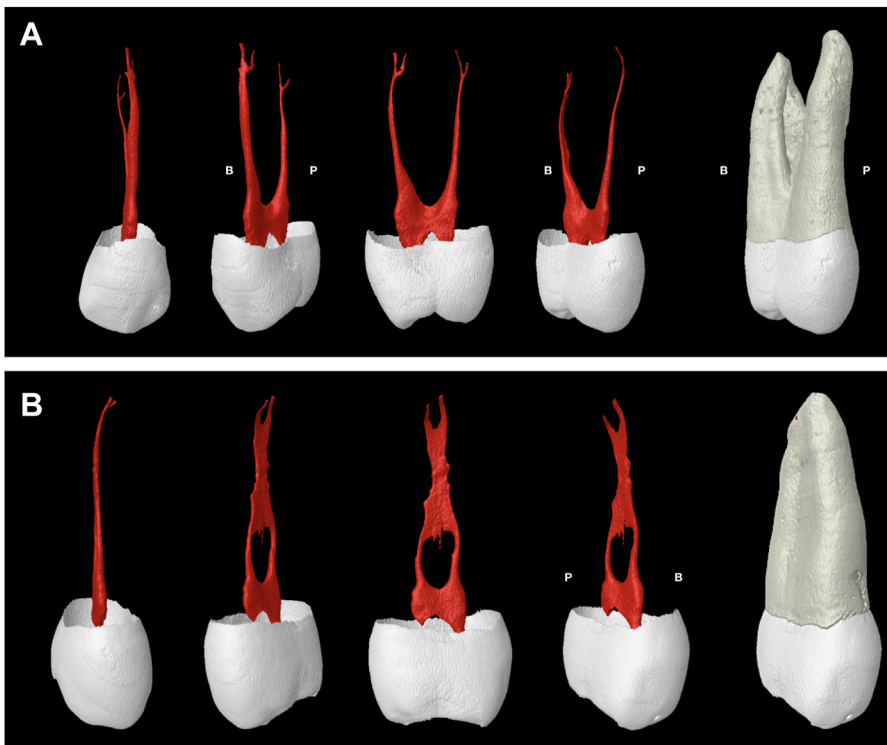


Fig. 1. Models of the maxillary premolars. (Top, A) Model of the maxillary first premolar. Note the presence of a buccal (B) and a palatal independent root (P). Each root presents 2 foramina at the apical third. (Bottom, B) Model of the maxillary second premolar. This single rooted tooth can present 2 canals: one located in the buccal (B) and a second in the palatal side (P). Both canals join at the mid-root level, and the canal system then presents 2 independent foramina. *Courtesy of Dr Marco Versiani.*

canal, commonly referred to as MB2. Due to its anatomic complexity, MB roots of maxillary molars could be associated with a high percentage of untreated canals and apical inflammation compared to maxillary premolars after root canal treatment⁴ (Fig. 2). The maxillary second molars serve to complement the first molar in function. Its roots usually contain 3 or 4 canals and are typically equal to or slightly longer than those of the first molar.^{5,6} Although the maxillary second molars anatomy has

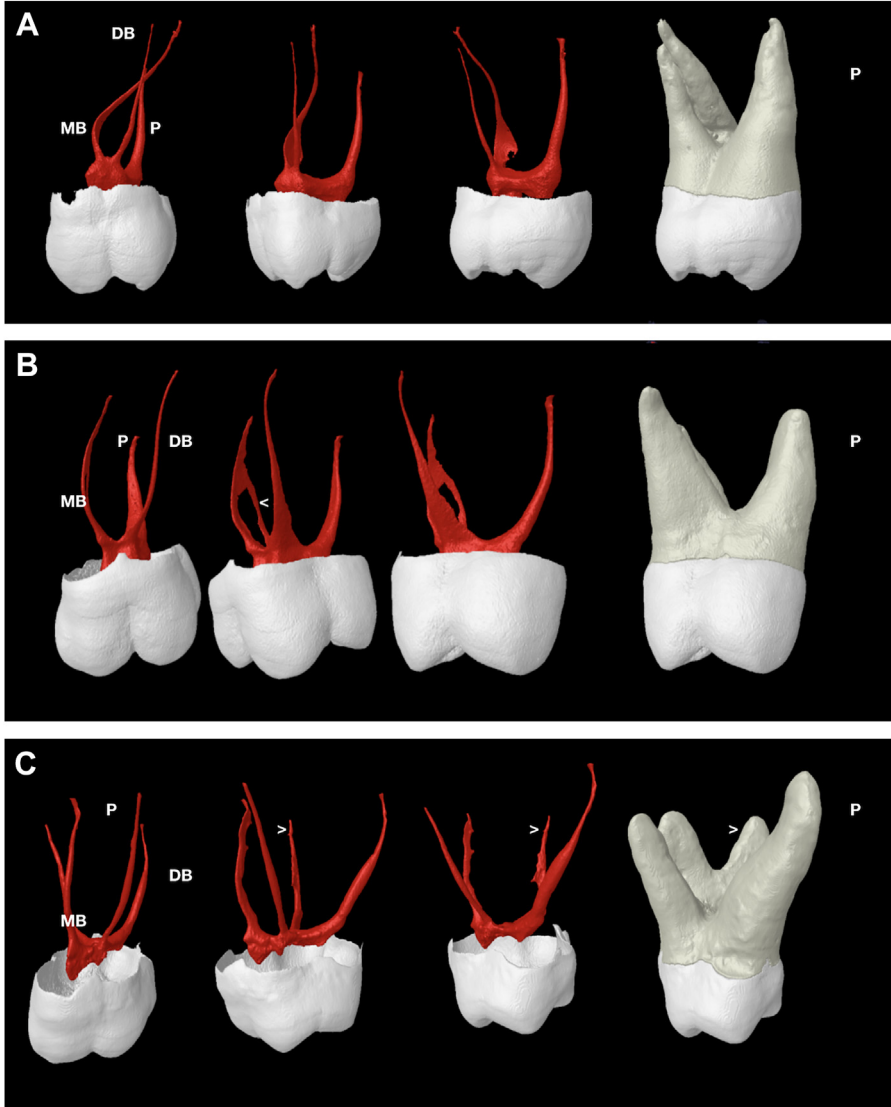


Fig. 2. Maxillary molar anatomy. (Top, A) Three roots and 3 root canals are present, the mesiobuccal (MB), Distobuccal (DB), and Palatal (P) root canals. Note the severe root canal curvature of the MB root. (Middle, B) Maxillary molar with 4 canals. An MB2 canal (<) can be observed in the MB root. Both canals join at the middle third, and the MB root presents a single foramen. (Bottom, C) A 4-rooted maxillary molar. The accessory root is depicted with (>). *Courtesy of Dr Marco Versiani.*

similarities with the maxillary first molar, the distinctive characteristic of this tooth is that its roots are grouped closer together or are occasionally fused.⁷ Fused roots can be challenging because they can present isthmuses or additional canals connecting some (or all) of the roots.^{8,9} The prevalence of fused roots in maxillary second molars ranges from 9.0% to 42.25%.^{8,10} The prevalence of fused roots in maxillary second molars is 4 times higher than with maxillary first molars¹¹ (Fig. 3).

In the maxillary dental arch, second molars are closest to the maxillary sinus, followed by first molars, and premolars.^{12,13} On average, the distance between the roots of maxillary molars and premolars and the maxillary sinus is approximately 1.97 mm, with the possibility of root tips protruding into the sinus floor, leading to minor elevations or prominences in the maxillary sinus mucosa (Schneiderian membrane)^{14,15} (Fig. 4). Therefore, maxillary molars and premolars are important in the context of ODS because the anatomic proximity of their roots to the maxillary sinus floor promotes endodontic infectious spread from the from the pulp chamber and periapical tissues into the maxillary sinus.¹⁶

ACCESSORY CANALS: APICAL RAMIFICATIONS, LATERAL CANALS, AND FURCATION CANALS

The dental literature has extensively discussed the anatomic intricacies of the root canal system, especially in the apical thirds. This region is marked by highly complex and variable microanatomy.^{7,17} These complexities include apical ramifications, lateral canals, subtle variations in foramina topography, and the potential pathologic effects from apical external root resorption and canal calcifications.¹⁸

The root canal refers to the segment of the pulp canal space located within the root. Accessory canals are small branches that connect the main pulp canal space with the periodontal ligament (PDL).^{17,18} These include lateral and furcation canals, and apical ramifications.¹⁹ Lateral canals are accessory canals found in the coronal or middle portion of the root, typically extending horizontally from the main canal.^{17,18,20} Furcation canals, on the other hand, are accessory canals located in the furcation area.¹⁷ These communications are formed during tooth development when periodontal

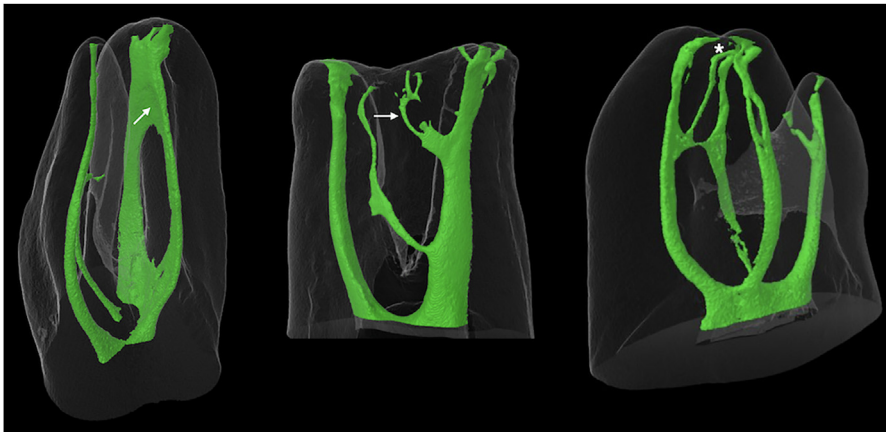


Fig. 3. Maxillary second molars with fused roots showing complex anatomic patterns including the presence of an isthmus (arrow in left image), an accessory canal (arrow in middle figure), and an apical delta (* in right image). Courtesy of Dr Ronald Ordinola-Zapata. Reprinted with permission, International Endodontic Journal 2017; 50:1192-1200.

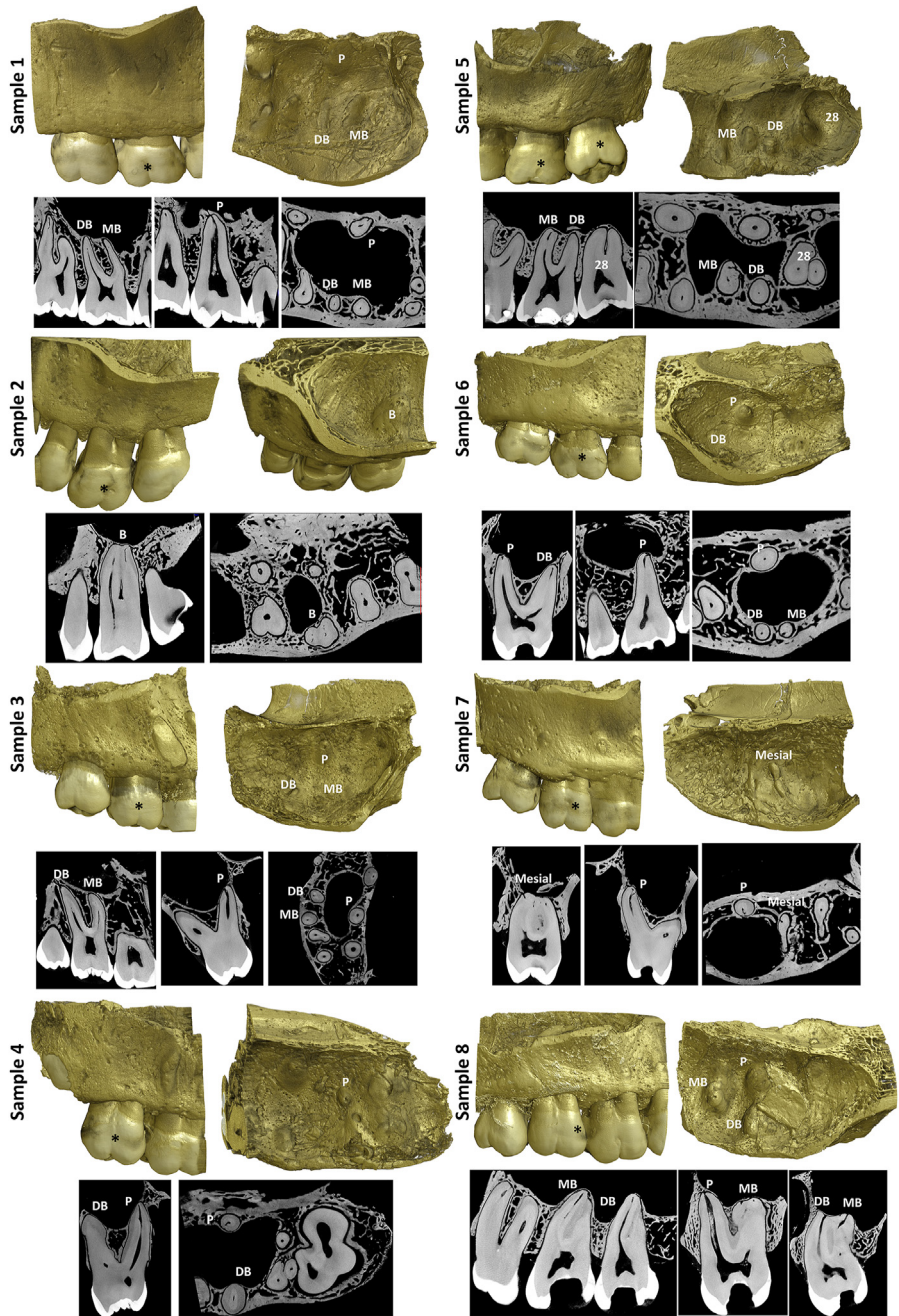


Fig. 4. Micro-computed tomography (CT) 3D models of 8 maxillary bone blocks (highlighted in yellow) conducted under a research protocol approved by the local Forensic Department and the National Committee on Health Research Ethics (protocol #931.732). The models illustrate the protrusion of maxillary molar roots into the maxillary sinus floor. Each model is accompanied by 2-Dimensional (2-D) tomographic projections (coronal, axial, and sagittal)

vessels are trapped during the fusion of the diaphragm, which becomes the floor of the pulp chamber.²¹ Apical ramifications occur when the main canal branches into multiple accessory canals near the root apex.¹⁷ This results in a complex root canal system characterized by an intricate network of ramifications in close proximity to the PDL.²² (see Fig. 3).

Accessory canals may not be readily apparent in periapical radiographs, but their presence may be inferred by the presence of a lateral root lesion or periodontal inflammation with bone resorption associated with concurrent pulp necrosis ("endo-perio lesion"). Lateral canals can be visualized on CBCT scans if the voxel size is small enough to reveal them.²³ In the context of pulp necrosis and apical periodontitis, accessory canals are pathways connecting the main root canal with the periodontal tissues. This allows for spread of bacteria and their byproducts into this area, leading to the development of furcation and lateral root lesions.²⁴ De Deus²⁵ conducted a study on 1140 teeth that revealed 27.4% of teeth exhibiting accessory canals, primarily located at the apical area (17%) followed by the middle (8.8%) and coronal (1.6%) thirds. In another study of 2400 teeth, Vertucci²⁶ demonstrated ramifications to be most common in the apical region (73.5%) compared to middle (11.4%) and coronal (6.3%) thirds.²⁶ A morphologic investigation of 100 molars revealed that nearly 80% of the specimens contained lateral and accessory foramina, with diameters ranging from 10 to 200 μm .²⁷ These dimensions are noticeably smaller than those of the main canal but larger than dentinal tubules.²⁸ The size of these ramifications could influence the accumulation of bacterial by-products, with larger diameters being more conducive to infectious spread.²⁹ The largest diameter recorded for an accessory foramen is approximately 2 to 3 times smaller than the average diameters reported for the main apical foramen.^{30,31}

The apical foramen serves as the main point of entry of vascularization from the PDL into the pulp canal space.²⁰ It contains neural, vascular, and connective tissues.^{30,32} Although the major apical foramen and the anatomic root apex are in close proximity, they rarely align precisely.³³ The distance can vary based on age or tooth type, but typically ranges from 0.2 to 3.0 mm.^{7,34} The foramen's shape may be round, oval, elliptical, or semilunar. Additionally, the location and shape of the fully formed foramen can differ not only between teeth but also within the same tooth at different stages of life.⁷

Studies have shown that maxillary premolars and the MB roots of maxillary molars exhibit the highest rates of harboring multiple apical foramina³⁵ ranging from 1 to 16.³⁶ In a study involving 493 human teeth,²⁴ the authors observed a high prevalence of lateral canals and apical ramifications in maxillary premolars (79.8%) and molars (55.8%). Similarly, Gao and colleagues³⁷ assessed the apical morphology of 1400 extracted teeth and found apical ramifications in 15% of maxillary premolars and 10.5% of molars.

In case of pulp necrosis, if no treatment is provided, there is a high likelihood that the apical tissue including ramifications will become infected over time. The complexities in the apical third of the root canal can interfere with infection control since antimicrobials may not diffuse well into this intricate microanatomic environment. Root canal treatment requires the use of specialized equipment and training to access and treat



of the relevant teeth. The 3-D reconstructions and micro-CT slides reveal the sinus floor morphology and the close proximity of the maxillary molar roots with the maxillary sinus. MB: mesiobuccal root; DB: distobuccal root; P: palatal root; B: buccal roots (fused); 28: maxillary third left molar. (Courtesy of Dr Marco Versiani.)

the apical third of these teeth. Consequently, both the complexity of the root canal anatomy in the apical third, and the preoperative presence of pulp necrosis with apical periodontitis are recognized as significant factors influencing the success rate of endodontic treatment.²⁹ Failure to eliminate the microbial burden in the pulp space during root canal treatment may account for an unresolved apical periodontitis that develops into ODS.

ENDODONTIC INFECTIONS

The dental pulp consists of a well-vascularized and innervated connective tissue, surrounded by enamel and dentin.³² Like other connective tissues, the dental pulp exhibits antimicrobial defenses against oral bacteria. The most common pathways for bacterial invasion are caries, dental trauma, cracked teeth, and iatrogenic procedures. Without dental restoration or root canal treatment, colonization of the root canal space by oral bacteria ensues. Left untreated, pulpal bacterial invasion may then progress to cause inflammatory or infectious apical periodontitis. **Fig. 5** shows microscopic relationships of infection spreading between the dentin and the root canal space.

Apical periodontitis begins as an inflammatory response in the apical/periapical tissues to underlying pulpal infection. Pro-inflammatory cytokines such as interleukin-1, IL-6, and tumor necrosis factor- α are elevated in endodontic infections and promote osteoclastic activity,³⁸ leading to characteristic bone erosion and periapical radiolucencies on imaging. Of note, bone resorption in the setting of apical periodontitis represents an adaptive and protective role. This is supported by observations showing that bacteria are often not found in most of the apical tissues in these cases, and that osteomyelitis is an infrequent complication of apical periodontitis.³⁹ However, these conditions can progress to become periapical abscesses, which can lead to ODS.

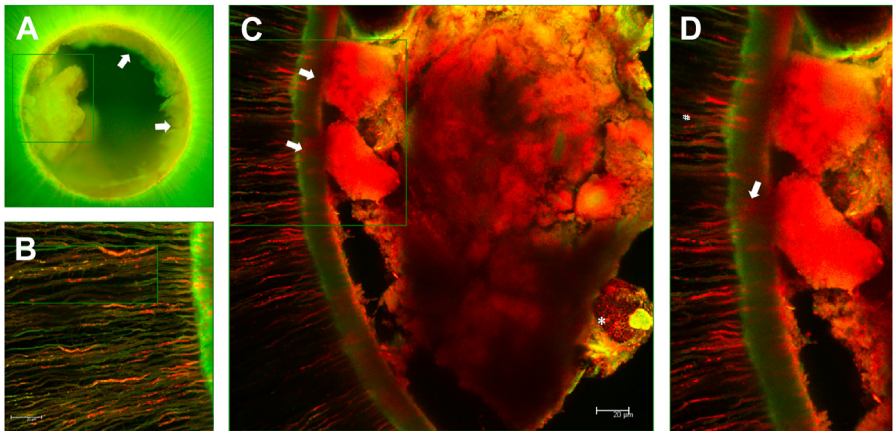


Fig. 5. Root canal space showing pulp necrosis. The sample was stained using the acridine orange dye; the microbial RNA is usually stained in red/orange. (A) Panoramic view of the root canal content showing a large mass of necrotic tissue and organic structures attached to root canal walls (arrows). (B) Magnification of the dentin showing a dense infection within dentinal tubules (orange fillings in the tubules). (C) Biofilm/dentin interface. A complex microbial amorphous structure heavily stained, attached to the root canal wall (arrows in green square). (D) Magnification of the dentin/biofilm interface (shown in c) revealing microbial cells invading the predentin and several dentinal tubules (arrow). Courtesy of Dr Ronald Ordinola-Zapata, reprinted with permission. "Root canal anatomy: Implications in Biofilm Disinfection" In: The Root Canal Biofilm, Chavez de Paz et al. Springer Ed.

Further details on endodontic infections can be found in the Dental Pathophysiology of Odontogenic Sinusitis: Endodontic infections article from this issue.

THE PERIODONTIUM

The periodontium refers to the anatomic structures involved in supporting the teeth in the jaw and consists of root cementum, the PDL, alveolar bone, and the gingiva.^{32,40} The PDL is a dense fibrous connective tissue with specific arrangements of Type I collagen fibers that anchor the tooth between the 2 mineralized abutments of root cementum and alveolar bone.³² The PDL allows physiologic movement of the tooth within the alveolar socket during mastication to safeguard against tooth damage from excessive masticatory forces. The PDL supplies nutrients to the cementum and surrounding bone through blood vessels and its innervation provides proprioception and tactile sensitivity.

The gingiva protects the underlying attachment apparatus from mechanical damage and entry of microorganisms from the oral environment. The gingival region of the periodontium supporting the cervical portion of the root is referred to as the 'marginal' periodontium. Accumulation of bacterial plaque around this gingival margin can stress the host defenses causing inflammation (gingivitis) and lead to periodontitis or loss of the marginal attachment and supporting crestal bone.

The apical and lateral mid-root regions of the periodontium are also susceptible to bacterial destruction from endodontic pathogens within the root canal space and their inflammatory by-products that communicate with the periodontium through apical and lateral canal foramina.²⁴ Destruction of this sub-gingival periodontium by endodontic bacterial pathogens and their inflammatory by-products are seen radiographically as localized widening of the PDL space surrounding canal foramina, and ultimately as radiolucent apical and lateral root lesions.

Lateral endodontic lesions that extend or progress gingivally into the marginal periodontium or root furcation may be easily mistaken on imaging as periodontal disease rather than endodontic disease. These endo-perio lesions are important to understand because these are usually caused by endodontic bacterial pathogens, and require a different treatment protocol than marginal periodontitis.

Pathologic damage to the dental periodontium, whether via endodontic or periodontal bacterial infections, has been shown to produce inflammatory edema in maxillary sinus mucosa even with intact alveolar bone (without direct sinus mucosal contact). However, direct communication between the periodontium and sinus mucoperiosteum is not infrequent anatomically, providing a direct pathway for endodontic (apical periodontitis with or without periapical abscesses) and potentially periodontal infections (advanced periodontitis) to spread into the maxillary sinus to cause ODS.

POSTERIOR SUPERIOR ALVEOLAR ARTERY AND PROCEDURAL IMPLICATIONS

The anatomic relationship of the posterior superior alveolar artery (PSAA) to the apical portions of the posterior maxillary teeth holds critical importance for the planning and execution of certain dental surgical procedures like root-end resections during apicoectomy, and maxillary sinus augmentation (sinus lift) surgeries (**Fig. 6**). The PSAA primarily supplies blood to the maxillary sinus bone and mucosa and the posterior maxillary teeth. It originates from the internal maxillary artery, which courses through the infratemporal fossa and enters the maxillary sinus through the infratemporal surface of the maxilla.⁴¹

After failed root canal treatment, apicoectomy may be necessary. This surgery involves removing the apical portion(s) of a tooth's root(s) plus reverse root-filling to treat

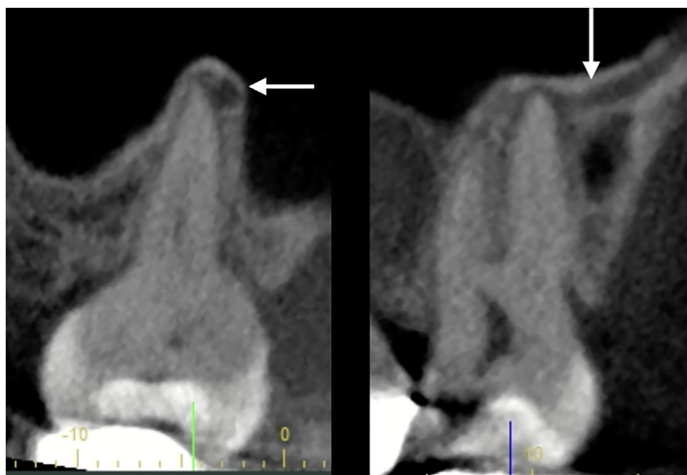


Fig. 6. Posterior superior alveolar artery (PSAA) artery (arrow) and its relationship with the apical third of maxillary molar and the maxillary sinus.

persistent apical/periapical pathology. This could include isolating and removing 1 or more buccal or palatal roots in maxillary molars and premolars. The PSAA's relationship to the buccal roots mainly requires attention to avoid extensive bleeding and ensure complete removal of the root structure without compromising adjacent healthy tissues.⁴² The PSAA can also be at the level of the root apex, and can be confused with apical periodontitis since it may appear as periapical radiolucency on conventional radiographs. CBCT is the imaging modality of choice to evaluate the relationship between the sinus floor and vascular structures. One must also appreciate the location of the PSAA when performing maxillary augmentation surgery, as it can be a source of significant intraoperative bleeding.

CLINICS CARE POINTS

- Maxillary endodontic and periodontal dental anatomy is intricate, and understanding these anatomic regions is critical toward understanding the spread of dental infections causing ODS, especially from the posterior maxillary teeth.
- Appreciating the complex root canal anatomy of posterior maxillary dentition is also essential if performing root canal treatments, as missed or untreated canals will lead to treatment failures and potentially further progression of ODS.
- All patients with periodontal lesions and ODS should undergo pulp vitality testing to determine whether the source is endodontic. Advanced periodontitis is an uncommon cause of ODS, but understanding the relationships of the periodontal structures is helpful in understanding both periodontal and endo-periodontal lesions that may cause ODS.

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